

Technology News

Batteries that charge within seconds

Lithium-ion batteries are the most common rechargeable batteries found in consumer electronics products. However, these take a long time to charge. A new manufacturing method has been discovered that could lead to smaller, lighter lithium-ion batteries which can be charged in just seconds. The approach only requires simple changes to the production process of a well-known material.

Until now, it was thought that there was a speed limit on the lithium ions and electrons that pass through the batteries to form an electrochemical circuit. Gerbrand Ceder, at the Massachusetts Institute of Technology (MIT), USA, and his colleagues have used a computer simulation to model the movements of ions and electrons in a variant of the standard lithium material known as 'lithium iron phosphate.' The simulation indicated that ions were moving at great speed. The problem turned out to be the way ions passed through the material.

The ions pass through minuscule tunnels, whose entrances are present at the surface of the material. The team discovered that to get into these channels, the ions had to be positioned directly in front of the tunnel entrances. The solution, Ceder discovered, was to engineer the material such that it has a so-called 'beltway' that guides the ions towards the tunnel entrances.

A prototype battery made using this technique could be charged in less than 20 seconds—in comparison to six minutes with an untreated sample of the material.

Affordable solar panels for homes

One of the biggest problems with solar panels is their high cost. Their price can easily exceed \$30,000 to outfit a



Veranda's flower-shaped, affordable solar panels

residential roof. For people who typically have smaller budgets but aspire to live greener lifestyles, California-based Veranda Solar has created a starter solar system that expands as your budget does. The company plans to sell these panels at roughly \$600 each later this year.

The Veranda panels snap together, so people will be able to buy just one to start and add more later on if they like. The solar inverter, which converts the direct current (DC) electricity from the panels into alternating current (AC) electricity that can be used in the electric grid, plugs right into a wall socket.

So far, the solar industry has stuck with a rectangular form factor. Veranda's flower-shaped panels will have a very different form factor from what people are used to.

Flexible solar cells printed like money

In a trial near Melbourne, printing presses normally used to make Australian dollar bills produced solar power cells. The machines arranged and stamped flexible solar panels onto a plastic film.

The cells so produced were only 3 per cent efficient, meaning they could convert only a small amount of

solar energy into electricity. But Commonwealth Scientific and Industrial Research Organisation (CSIRO) project leader Gerry Wilson expects the output to more than double by next year and top 10 per cent after that. He says that the solar cells would be ready for mass production in five years.

The main advantage of this method is that the cells can now be produced in vast sheets or rolls, making them ideal for windows or large-scale, rooftop applications. The printers could turn out 100 kilometres of solar sheet every day, according to CSIRO.

Wireless electricity

This year's *Consumer Electronics Show* saw the launch of gadgets with pads that allow wireless charging. The pads work via a coil built into them that generates a magnetic field when plugged into an outlet. If you place a gadget with its own coil built into it on the pad, the magnetic field generates a current within the second coil, in turn charging the device.

Palm's Pre features such a pad. WildCharge has built pads that work with any phone as long as you attach their adaptor to your mobile.

Though wireless charging is attractive, the only way it can be adopted fully is if multiple items are charged